



Research Basics: From the Introduction to the Research Questions

GRAD 695

In this lecture, we will cover the full scope of what you need to include in your proposal and how it all fits together. And then starting next week, we'll start talking about specific types of research design and the various ways you might go about collecting your data.



What is an Introduction?

- The first page (-ish) of a research paper
- Sets up the context and importance of the topic
- End with a goal statement about the project

The first thing you're going to write is an introduction. At this point you should have at least skimmed some research articles and seen the general format of them, and after the abstract (which is often the last part written; you will write yours in 699), the first part is an introduction though it might not be labeled as such. There's just different norms and expectations in different fields and for different publications for what labels and headers are used. Often the introduction is just the first few paragraphs of the paper before it gets to the literature review.

For the research proposal for this class, I'm not expecting anything more than about a page for an introduction, because sometimes this can be as short as a single paragraph, but more commonly 2 to 3 paragraphs. The purpose of the introduction section is to really set up what the problem is, set up the context and the importance or the implications for what it is you're studying.

So this is a kind of high level overview of what you're studying and why, and at the end of the introduction you should end with something like a goal statement about what's the big thing, the broad purpose of the work, What is it you're trying to either build or trying to learn?

Parts of an Introduction

Narrative hook

Background on the topic

Importance of the topic

Goal Statement

In terms of the parts of the introduction, there are four basic components. Don't use subheadings in the introduction, but rather in the two to three paragraphs you write, you need to hit all of these topics, or all of these functions.

Narrative Hook

- Start your introduction with something to grab your reader's attention and make them want to keep reading
- Potential Hooks:
 - Statistics
 - Anecdote or story
 - Interesting finding
 - Real world event or phenomena
- Note: The introduction is where you can use a few non-academic sources (e.g. news articles, reputable blogs, social media, etc)

One component to have some sort of narrative hook.

This is usually in the first few sentences, but it could be scattered throughout the introduction.

This is information that gets the reader's attention and makes it clear why what you're studying, what you're writing about is important.

This could be starting off with some statistics of how many people does this problem affect or how big is the problem in terms of money or time?

The other thing you can start with is anecdotes or stories, anecdotes generally not good in science.

They do not really constitute evidence, but stories can be good inspirations for scientific work.

So, a news story or some other current event real world event that is relevant to the problem we're studying and that gives the reader an example of what this problem looks like, What's the real world application of it? Even if it's just a one off anecdote.

So you're starting with something interesting that really allows you to show and contextualize and make the problem concrete.

This is pretty much the only place in the whole research article where you can

have non academic sources.

So unlike once you get to the literature review where you should be pulling primarily from research articles and other kind of academic peer reviewed work. In the introduction you might even cite social media post or news articles or blogs. Avoid problematic or fake sources but you have a lot more flexibility in this section in terms of the types of citations you have.

And note that you do need to cite these things, so if you're referencing a real world phenomenon, you need to cite a news article or some source that you know gives evidence for that thing happening, or how people talk about it. Or if you're citing statistics you need to find a Gallup or Pew poll or whoever calculated those statistics to give support and evidence for them.

Background on the Topic

- Provide basic background on the topic
- Define what it is you want to study
- Orient your reader to any special terminology and the context of the topic

Another component is the basic background of the topic, so whatever it is you you're studying,

What are the boundaries?

What are the important terms?

Maybe the context?

What does the thing you're studying look like?

How long has it been studied?

How do people generally talk about it, approach it?

This doesn't have to be long; can be a few sentences, but just an orientation to what context we're working with.

Importance of the Topic

- Explicitly state why the topic is important
- If you study this topic, what could be learned or how could the world be better?
- Give your reader a reason to care about the project

You're also using this to show the importance of the topic.

Why should anyone care about this?

What are the real world applications?

What are the implications?

If this problem is solved, what does that mean?

Or what that could that mean in terms of money saved or life saved or time saved whatever?

Just whatever is relevant to your problem to show that it is more than a simple academic problem or a thought exercise.

So these first three things, those should all happen somewhere in this, two to three paragraphs you're writing to set up the introduction then you want to end the introduction with a goal statement.

A Goal Statement

- Broad statement about what you want to accomplish in your project
- Uses constructs not variables
- Think about what you want to know once you finish your paper
- Statement should be 'The goal of the current project is to...'

So basically, what's the broad thing you want to accomplish in your project? And you don't want to get overly specific yet, so don't cite specific variables you're going to study, but cite the concepts.

The general relationships or concepts that you're going to be studying. And what are you doing with them?

So this should let the reader know what it is you're going to discover or find by the end of the paper.

There are variations on how this works based on different author styles, but given that you're all beginning scientific writers.

It's useful to follow convention here, and so I want you to be very explicit and actually use the phrase. The goal of the current project is to and then fill in the blank. That just makes it completely explicit and clear that this is what it is you're hoping to accomplish with your project.

- The goal of the current project is to explore how COVID has impacted healthcare industry stocks.
- The goal of the study is to determine how political candidates use Twitter push specific messaging during campaigns.
- The goal of the current study is to examine how negative reviews of companies impact stock prices.

Example Goal Statements

And here's a few examples of how that might look.

So if you're trying to derive insights into something use words like explore or determine

If you're trying to understand relationships between variables, examine.

If you're trying to build a model or tool, you can say that you know the goal of the current study is to build X or develop X.

Whatever it is.

But this is the kind of general form it should take.

And this will end the introduction and set you up to transition to your literature review.

Note that the rubric for the assignment will list these components

But they're not just a sequence of things you talk about.

You shouldn't have a paragraph on the background. A paragraph on the importance and an end on your goal statement,

Rather they should be integrated together.

One thing I suggest to help is, as you're reading research articles for your literature review.

Consider how they wrote their introductions, the information they provide and the flow of ideas. If you find really good introduction that you think is a good introduction for that paper, try to model yours after that in terms of the flow and the structure.



Literature Review

Connects the problem and general goal to your specific approach to it

Meant to show what has been done and what the accepted approaches are

Provides support and justification to your methodological and analytic approach

Leads to the purpose of the study and hypotheses/research questions

So once you have your problem kind of set up and got everyone on the same page in terms of what you're doing, the next section of any research article or any research proposal is the literature review (sometimes called a related works section).

So this is where you connect the general problem and goal to specifically, how are you approaching it?

What specific research questions are you answering? What specific hypothesis?

The literature review will go from starting broad and explaining what's known in this field. What's already been done. What are the things everyone agrees on. And then you'll get more and more specific to doing, maybe critiques of past work or showing what the limitations or the gaps of specific studies are and narrowed down to till you get to the purpose of your study, the specific thing you're trying to do as well as your research questions and through this the literature review should support and justify why you're going to study this problem in this way.

Why are you choosing this research design?

Why are you choosing this analytic approach?

The literature review should end on a paragraph that gives the kind of general approach and the purpose as well as the research questions.

This can be just a few sentences followed by the research questions, but you want to specify your methodological approach to the problems.

Main Purposes of any Scientific Study

- Purpose is more specific than the goal of the project and should reflect the research design to be used
- ‘The purpose of this study is to describe...’
(focus on gathering and summarizing information on a topic)
- ‘The purpose of this study is to explore...’
(focus on generating or discovering new information about a topic)
- ‘The purpose of this study is to explain...’
(focus on determining the causes or relationships between variables)
- ‘The purpose of this study is to predict...’
(focus on prediction and forecasting some outcome)

Again, you can use this kind of formulaic start to help you frame everything. These are the four versions you can have.

If you're summarizing or trying or doing descriptive work, you might say something like purpose of the study is to describe and then whatever that is.

If you're doing exploratory work, that's also fine.

If you're trying to explain complex relationships, use the phrase to explain, or

If you're building predictive models or trying to generate predictions or forecast to predict

If you can also say the purpose of the study is to build a model, develop an algorithm that does X.

So you have this and then you have either your hypothesis or your research questions.

Hypothesis Testing

- Recent source of controversy in science
- At a basic level, hypothesis testing is simply defining all the possible outcomes of your study
- The academic intervention does not have an effect on grades.
- The academic intervention has a negative effect on grades.
- The academic intervention has a positive effect on grades.
- In addition to 'statistical significance', important to consider:
 - Effect size
 - Practical significance

Historically, our traditionally speaking academic research has been very focused on hypothesis testing.

That is, researchers make predictions about the relationship between variables, and then they test it with inferential statistics.

Now that was fine back in the day when you had to do math by hand and your sample sizes were small and when the work was very theoretically driven where researchers developed models and frameworks that they could conduct studies to support or not the predictions from that framework.

Nowadays it doesn't make as much sense for a few different reasons.

One, it was never ideal; It was always a work around to the limitations of computational methods and limitations of data collection methods and was based on making a lot of assumptions about data and about data collection.

So this is where you get the assumptions for regression. You know that your variables have to be normally distributed, that there has to be independence of data points. The errors have to be distributed normally. etc

Many data sets that meet those assumptions and so in the last decade or so, there's been a shift away from all research being hypothesis testing to kind of a general discussion debate over what.

What the alternatives are and what the best approach to actual science?

So the with hypothesis testing is used if your whole study is based on some theories of past work that say you know this is how these variables relate to each other, and this is like what the if you collect data.

And while hypothesis testing basically relies on P values to make judgments, which is not great because statistical significance does not mean practical significance and there's better measures for the magnitude of effects.

Plus, the words statistically significant are improperly used most of the time.

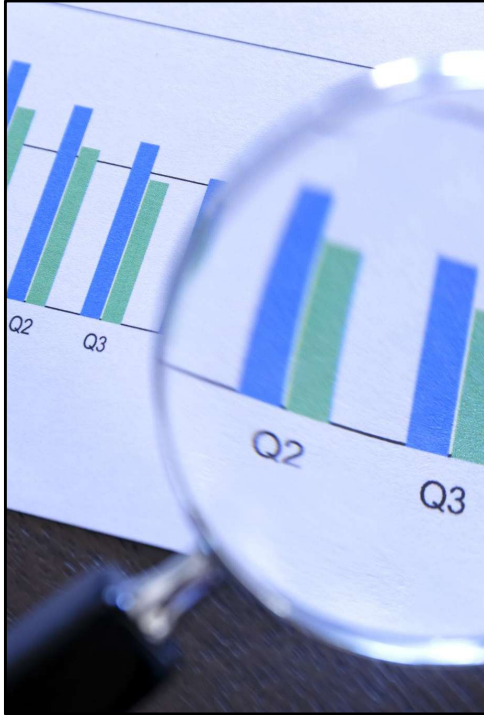
So you want to make sure that you're not just focusing on finding p value less than .05 because you can always do that if you throw enough data at it. Basically, with a large enough sample size, anything statistically significant, you want to consider the practical significance, ie the effect size to measure the magnitude of the effect. The difference the prediction, but also in real terms what does it mean?

How much money does this save?

How many lives does this save?

What does this mean in terms of how people actually interact with the world.?

So even if you're doing hypothesis testing, you should consider those things and consider how you can measure the real world application beyond just the P values.



Hypotheses

- Single, simple statement predicting the outcome of your study
- Testable with a statistical analysis
- All variables quantitatively defined
- Population is defined
- Prediction should be supported by the literature review

The other issue with hypothesis, if you're going to do that is again, they need to be supported by the literature review.

So you need to be able to show specifically what is the past theories and frameworks you're leaning on to say that these are testable predictions. And then two hypotheses rely on you to clearly define what your population is. What is the kind of realm?

The sample space in which you're conducting research.

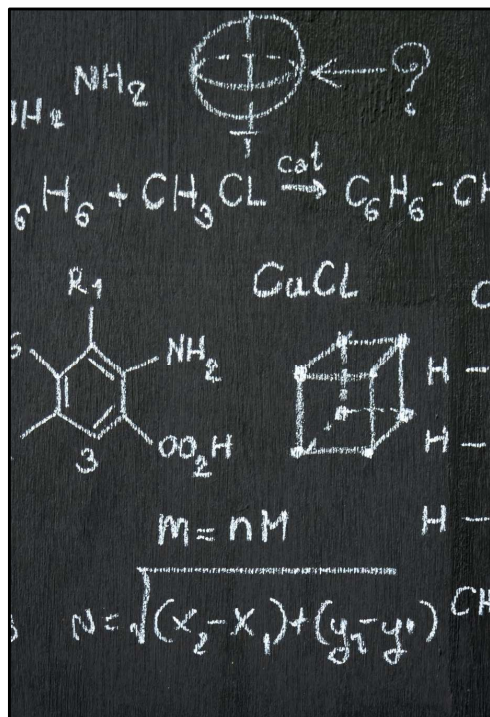
And what are the outcomes and the variables that you're going to model? All of these have to be testable with statistical analysis.

So basically a proper hypothesis. You should be able to look at the hypotheses, look at the output of a statistical test. So like a t-test or regression, what have you and say yes or no.

Yes, this hypothesis is supported by the data.

No, this hypothesis is not supported by the data.

And if you can't do that, then your hypotheses are not well defined, and it's possible that hypotheses just don't make sense for the type of problem that you're addressing.



Null versus Alternative Hypotheses

- An alternative hypothesis is the result you expect/hope to happen.
- The null hypothesis is the other result.
- Statistically, these are denoted as H_0 and H_1 and can be expressed mathematically.
 - $H_0: M_1 = M_2$
 - $H_1: M_1 \neq M_2$
- Can be directional or non-directional but must cover all possible outcomes.
- In a research report, generally only the alternative hypothesis is explicitly defined and is described in words rather than mathematically
 - Positive mentions of a stock on Twitter increase stock price for the company. (testable with the right data and a simple regression model)

In research reports or in research articles, if you've been reading any that include hypothesis, you'll note that generally they don't actually state what the null hypothesis is, they only state what the alternative hypothesis is, and the null hypothesis is just implied.

Which causes some issues because if it's not well defined, alternative hypothesis, you're not really sure what the comparison being made is. So for all of these reasons and more, I don't actually recommend you necessarily framing your research in terms of hypothesis.

- The number of retweets of a political candidate's tweets is positively correlated with their approval rating.
- The use of Instagram for marketing increases a small business's profits.
- The consumer confidence index in one quarter significantly predicts the Dow index in the following quarter.

Example Hypotheses

Research Questions

- Alternative to defining a hypothesis
- Functionally similar to a hypothesis but written as a question rather than a prediction
- Still must be testable with the statistical analysis leading to a clear answer
- Used when:
 - Specific prediction not supported by past research such as in exploratory studies
 - Hypothesis testing will not be used (e.g. prediction models, descriptive studies)

Better is research questions, so this is an alternative to defining the hypothesis. It is functionally the same as a hypothesis, but rather than approach research as being a prediction that you're making with a yes or no response.

You frame it as, What do you learn about this question from this data?

It still has to be testable and you still have to phrase it or construct it in such a way that you can answer the question with some sort of analytic model.

But it's not so specific that there's only one way to look at it.

So if you're doing prediction models or you're doing descriptive studies, research questions will give you the freedom to say or to look at the data in a few different ways and not rely on P values and this kind of yes or no. Something is or is not supported.

Characteristics of Good Research Questions

- Clearly states target population
 - What/who do you want your findings to generalize to?
- Descriptive
 - Clearly states the variables to be studied
- Adds new knowledge to existing knowledge
 - Addresses gap or limitation in past work
- Scope of data needed is clear
 - Know what data needed and how to acquire
- Clear objective
 - What will you know after doing this study?

Now ideally you would still define your target population here, What are the boundaries of the situations/people/the whatever that you're finding should generalize to or that you're pulling your data from?

It needs to still state what are the variables to be studied, what kind of data are you working with and how is that data capturing different features or different variables?

In this section, you're also going to discuss, you know, what are you adding on to the existing knowledge?

This whole section here at the end of the literature review, you know, starts with the purpose and then you talk a little bit about these things in order to support your question.

You might say that you know past work has looked at this, but they haven't considered this other variable.

You have a paragraph or so describing what it is you're trying to look for and find and kind of what the scope and the boundaries are and the objectives are and that all supports a clear question or questions.

You can have more than one research question.

- Can a neural network based model outperform traditional models for forecasting stock prices?
- What economic and social factors, such as GDP, education levels, and overall health indicators, explain a country's subjective wellbeing?
- How does the content of tweets impact the Tweeters' influence (e.g. number of followers and retweets)?

Example Research Questions

And these are just some examples of good research questions. One common mistake I see is a lot of students will have a research question.

Something to the effect.

Can X predict Y?

That's a bad research question of course X can predict Y.

That's not what you actually want to know, you want to know can X predict Y better than Z?

Or can this model outperform traditional model?

Or does these set of features do better than chance at predicting something?

So you want to make it clear in your research question what are you actually comparing?

What do you want to learn from the data compared to what's already been done from your literature review?

I would suggest most of you trying to frame your work as a research question cause not only did most of you are working on problems that are more data-driven rather than theoretically driven, but this also gives you a bit more flexibility as you design your work to make adjustments and shift variables if needed later on.

So at the end of your literature review, you should either have a hypothesis or have a research question.

Generally, you don't do both unless they're about two different things.

But you have to have at least one.



Building Blocks of Research Design

- Data
- Measurement
- Analysis

The final section draft you'll write is your proposed method section.

This is the part of the proposal where you basically lay out your plan of how are you going to answer that research question and ultimately solve your or achieve your initial goal for the project.

The structure and content of the research design will depend heavily on your methodological approach.

If you're doing kind of a more traditional social science approach to your problem by conducting interviews or collecting a survey data.

Then the structure of that method section is will look a lot different than if you are, you know, testing out parameters for AI model to see what yields the best performance.

But in all of them, no matter what the design is, you basically need to hit at least these three things.

What's the data? What data are using? How's it gonna be collected?

How are things measured? So what are your variables or your features? How are things quantified basically?

Then the analysis, How will you take the data and get insight? What either qualitative or quantitative models are you using on that data?

Data Quality

How accurate is the data?

- All data has measurement error
- Allowable level of error depends on the situation

Are the variables/labels in the data useful?

- The quantification of the data should match real-world behavior
- Multiple measures of the same variable can help

How well does the data capture the problem/situation in the real-world?

- Always consider how people approach the problem and how they will interact with the results

And a few things to keep in mind for each of these first data.

A lot of you will be working with preexisting open source data, so you're not necessarily going to be collecting data from scratch and are going to be depending on, you know, data that someone else collected.

And even if you are doing your own data collection, these are still applicable but how you address them is a little different.

So first thing is, how accurate is the data? All data has some level of measurement error even in you know tightly controlled clinical trials. There's some level of error involved. Lots of different ways that errors can be introduced into data.

How can you check for and quantify accuracy in the data?

Some of this might just be looking at distributions of variables to see are there are outliers.

Are there values that are weird? Like if one of your variables you have data set on people and you ask people their age and one of the values is 123; technically possible, but highly unlikely. Might represent an error.

If the values of the variables you're collecting have reasonable ranges, that's a way to check for error sometimes.

Another thing is are the variables. The labels in the data? Are they useful? Do they match up with real world behavior?

For example, in the realm of sentiment analysis, a very common NLP method, just widely applied to a lot of things, and most of the times it is doesn't on its own have any real world application. It doesn't necessarily tell you what you should do or what other people should do, or it doesn't give you any actionable insights. Especially if it's just positive or negative label. You need to know more about like why is a positive or negative.

What's the the strength of the positivity or negativity? Or you know, how does it relate to specific attributes or features?

This can apply in a lot of different cases because often the data that gets collected and recorded isn't necessarily the best way to quantify something. It's just the easiest way.

Which pros and cons? You know, sometimes the easier way to collect data is better than the right way that wouldn't have been feasible or would have had other problems, so there's trade-offs.

But the key here is to recognize when the data you have was collected was because it was easy, not because it was necessarily the best way to do it.

So that you can acknowledge that as a potentiall imitation and take that into account in doing your interpretation of the results.

Ideally you would have you know good measurements or have even multiple measures, multiple quantifications of the same variable, so that you can address some of the potential error and imprecision, but sometimes you have to work with what you have, and that's fine as long as you're honest and transparent about the limits of what you have.

And part of that is, you know, pretty much all of your studies going to be figuring out ways you can apply research to very real practical problems.

And so you always want to consider is the data you're working with a good representation to the problem, the processes in the real world is that how people interact.

This is evident in some of the work around AI; sometimes computer engineers, computer scientists, make assumptions about people that are not necessarily founded in evidence or good practice or psychology.

And the products, the systems they end up making might not actually be all that useful to the end users because they never really considered how people would interact with the chat bot or with the system.

So that's another thing to just keep in mind to help you both try to head off problems at the beginning, but also to more accurately and honestly interpret the

results at the end and have a better, more realistic sense of what the limitations of what you're doing are.

Sampling



Does the data represent the people/situations that the model/algorithm will be applied to?



How diverse is the data set?



Are there some categories or ranges of variables missing?



How flexible is the model to apply to new situations and datasets?

Now related to this is sampling as pretty much regardless of what you're doing you're not going to be working with your full population.

You're going to be working with a subset of data that is hopefully representative of the larger possible data you could pull from.

But realistically, it's probably not representative and so it's important to both quantify and consider how that impacts the results.

So if you're building a model, building an algorithm, building an AI agent from training data or even fine tuning it, you want to consider does the data that you're using for training. Does it actually represent the people, the situations, the processes that that, that algorithm that model will be applied to later?

If it's going to be, you know, applied to completely different situations, completely different data points, it will likely make some weird errors and possibly make some big mistakes.

This is how you get, AI models and you know, news stories about how AI is either racist or biased or discriminatory.

It's because no one really considered in the training data whether it was as diverse as the real world cases and situations where people will be up using that model.

So quantifying the diversity, the kind of coverage of the training data set is essential to interpreting and explaining the results from any sort of model.

Especially in predictive models you want to make sure that if there are categories or ranges of the variables that are missing that is considered in the interpretation. And I would say this is not just a problem for the modern issues of AI. This is kind of a long standing problem.

In psychology, often researcher want to understand the extremes of human behaviors like aggression or violent behavior

But then they would survey college undergraduates and try to draw conclusions about war and other, you know, extreme events from the hypothetical behavior of some freshmen in a psychology class.

And they're not really capturing the full range of human experience there, nor the full range of human personality. so it doesn't make the work necessarily bad, but it does make the interpretation bad because they try to extrapolate and beyond what the data could actually do.

And you want to consider that in your own work.

In a lot of the Retrospective studies using of open source data. Sampling is always going to be a big issue.

You can't necessarily control whether or not the sampling was random or how random it was, or whether or not there were systematic areas where groups or population samples of people that just got missed.

But describing the data in detail and describing how it was collected will at least help to contextualize the data and the data quality.

Types of Variables

Categorical

- Nominal
- Ordinal

Numeric

- Interval
- Ratio

Features

- Extracted from unstructured data
- Developed from structured data

This is more for later once you kind of get to the writing the methods section and this is almost a bit too specific as in terms of how you'll talk about the data and the variables in the data in your methods section.

But it is worth keeping in mind that you want to consider the types of variables you're working with, because they do have different constraints as to what you can do with them.

So from a statistical point, you can get really detailed and get into this whole nominal ordinal, interval, whatever. Basically the two big ones that matter is categorical versus numeric. Is your data in discrete buckets or is it some sort of continuous number?

That limits what you can learn and what you can do from an analytic approach. And there's a third option here.

If you are working either in a predictive space or working with unstructured data. So text, images, audio, et cetera.

You might be talking about how you're converting variables into feature sets. So how are you taking? Product reviews and quantifying them in, you know, numeric vectors through topic modeling. Or word embeddings or whatever method you're using.

Another layer you kind of need to add in your methods of you know how are you going to take this inherently unquantifiable data and quantify it.



Conceptual Definitions

Variable definition that explains in plain language what the variable is, what it is designed to capture, and why it matters

These generally will go in your Literature Review as your definitions should be supported with sources showing how other researchers define that variable

If a variable has more than one definition, you just need to justify why you chose the one you did (i.e. best match for the domain, most common definition, how it is measured in your data)

Should write at least a couple of sentences about your main variables

In your literature review and on a high level, you can talk about conceptual definitions of things.

So these are definitions of your variables or definitions of your features, which are just kind of the plain language.

Straightforward.

Discussion of what they are, what their designed to capture.

So this would be, you know, talking about.

Sentiment as a general evaluation towards a person, place or thing.

That would be a conceptual definition of sentiment.

And this is where you kind of.

Expand on in a you know couple sentences about what? What it is you're trying to capture in your data.

You know, connect it to the real world processes real world relationships.

Note that something some variables are going to have more than one definition because like sentiment actually does have more than one definition.

On one hand, it could be, you know, the general evaluation towards a person placing thing.

On another you could talk about it as.

A effective response to.

Different situations to get it towards a more emotional side of the aisle.

So.

If you're working with variables like that, that might have more than one definition.

One discuss that in the literature.

Review, you know, acknowledge that there's disagreement on how this is defined.

But then you pick the one that either is the best match for what you're doing are the most common, or just how it's measured in the data you happen to have.

While, like I said, acknowledging that there is some disagreement. So that's the the top level broad level where you start for variables.



Operational Definitions

Explains in more technical/statistical language how the variable will be quantified, what the values of the variables mean, and how the values should be interpreted

These will go in your Methods section in a subsection dedicated to the variables in your study

Variables can have multiple operational definitions which is especially useful for complex variable to show robustness

Should write at least one full sentence for each variable

In the method section, you're gonna give the operational definition.
So this is where you get into the kind of technical or statistical language of how are you actually quantifying this thing?
How are you taking sentiment and converting it into something that could be analyzed in a statistical model?
And if it is something with different values, what do those values mean?
How should those values be interpreted?
So with the sentiment example, maybe.
For this study, we're saying that sentiment will be measured using the Vader lexicon approach, which.
Scores text.
On positivity and negativity on a scale from -1 to one.
All right, Peter.
No Bader Zero 10.
There's a lot of different measurements you can get from there, so I can't keep them all straight, but you're actually going to tie it to the specific.
Way that that variable is converted into numbers and just like conceptual definitions, there can in there can be and there often are multiple operational

definitions for the same concept.

Like sentiment can be measured with the Vader lexicon.

Or it could be measured with a pre trained model.

Where you can infer positive, negative neutral labels.

Or it could be captured with other lexicon methods like Luke or Athen, or several others.

So.

Again, you can just choose one if there's one that makes more sense for your specific problem than others.

But if possible, and if you know logical, you should include as many of those operational definitions as make sense for your problem, because each operational definition is gonna involve some level of measurement error or some kind of weaknesses to it.

And so the more different quantification you can add in there, the better you're kind of triangulating the real value.

And again, this goes in the method section and you should still be able to write about one full sentence for every variable or every set of features that you're going to be using in your study.

Analysis



More on this in later lectures



Should be consistent with goal and data



More complex is not automatically better

Now the last thing about go on the method section is kind of the analytic plan. You know, how are you going to actually take your data and what are you going to do with it to get it ready for doing statistics next semester?

Now, more on this later, because the analysis is going to be an analytic plan, is very much tied to the specific method and approach you're using.

So as we get into different methods and approaches, we'll discuss you know how this looks.

Analytically.

But the key here is to make sure whatever plan you have that it's consistent with the goal consistent with the the data and the type of data you have. And note that more complex is not automatically better just because it can build an AI model doesn't mean you.

Should build an AI model.

Sometimes all you need is a logistic regression classifier.

So making sure that whatever you do, whatever choices you make for the analysis.

Are based in logic and sound science and not just. This is the trendiest thing right now.

Every decision here should be justified and should be justifiable.

Types of Variable Relationships

If the goal of your project is descriptive:

- Just define all your variables

If the goal of your project is explanatory:

- Define independent variables (or predictors) and dependent variables (or outcomes)

If the goal of your project is predictive:

- Define your features (or inputs) and your outcome (or output)

Now, in doing the analysis, this is also where you're gonna need to translate your the goal of your project to a specific analytic approach in a specific methodological approach.

So if the goal of your project is generally just descriptive or exploratory.

You're gonna just basically define the variables and discuss the data, and then discuss the various EDA methods you're gonna use.

To describe the dev.

If your goal is explanatory or you know you're trying to figure out relationships.

Between variables and you know model them. You're gonna need to make sure that you define kind of what role each variable is playing in your model.

So you want to separate your variables into independent variables or predictors and your dependent variables or outcomes to kind of.

Provide structure to the data and how it's gonna be used in the analysis.

Analysis.

If you're building predictive models, then you need to make sure that you have your features or your inputs defined and your outcome or your output defined.

That way it's clear you know what side of the analytic model is each you know component in your data going into.



Refinement

If your data cannot answer your research question,

- Change your research question
- Get different data
- Consider how data could be extract (e.g. feature engineering)

In the end, if the analysis does not clearly answer the research question, then you didn't design a good study